

Trainings ▾

Preparatory Steps

Automotive with CM870 and CM871

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



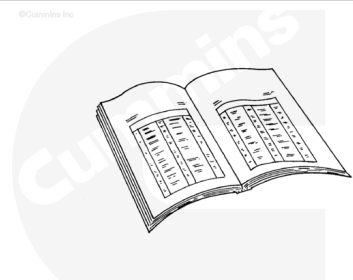
- Disconnect the battery cables. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/10/10-013-009.html)
- Remove the exhaust gas recirculation (EGR) mixer. Refer to Procedure 011-021 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-021-tr.html)
- Disconnect the air intake connection. Refer to Procedure 010-080 in Section 10.
(/qs3/pubsys2/xml/en/procedures/10/10-010-080-tr.html)
- Remove the engine wiring harness from all temperature and pressure sensors, fueling and timing actuators, the fuel shutoff valve, the water-in-fuel sensor, and the fuel lift pump.
- Remove the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-026.html)
- Remove the EGR differential pressure sensor. Use the following procedure in the Troubleshooting and Repair Manual, CM870 Electronic Control System, Signature™ and ISX Engines, Bulletin 4021334. Refer to Procedure 019-370 in Section 19.
(/qs3/pubsys2/xml/en/procedures/70/70-019-370.html)
- Remove the EGR differential pressure sensor adapter. Refer to Procedure 011-028 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-028.html)

- Remove the customer accessory bracket.
- Disconnect the fuel supply and fuel drain lines. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/10/10-006-024.html)
- Remove the lift pump. Use the following procedure in the Troubleshooting and Repair Manual, CM870 Electronic Control System, Signature™ and ISX Engines, Bulletin 4021334. Refer to Procedure 019-396 in Section 19.
(/qs3/pubsys2/xml/en/procedures/70/70-019-396.html)
- Remove the engine control module (ECM) cooling plate fuel supply line (bottom) from the integrated fuel system module (IFSM) and the ECM cooling plate fuel return line (top) from the fuel pump. Refer to Procedure 006-006 in Section 6. (/qs3/pubsys2/xml/en/procedures/10/10-006-006-tr.html)
- Remove the fuel filter. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/10/10-006-015-tr.html)
- Remove the accessory bracket located above the IFSM. Refer to Procedure 001-082 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-082.html)

CM570

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

- Disconnect the battery cables. Refer to Procedure 013-009 in Section 13.
(/qs3/pubsys2/xml/en/procedures/10/10-013-009.html)
- Remove the air inlet piping. Refer to Procedure 010-080 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-080-tr.html)
- Disconnect the engine wiring harness from all pressure and temperature sensors, fueling and timing actuators, the fuel shut off valve, the water-in-fuel sensor, and the fuel lift pump.
- Remove the fuel control covers.
- Disconnect the air intake connection. Refer to Procedure 010-080 in Section 10.

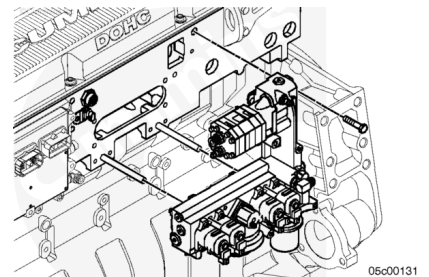
(/qs3/pubsys2/xml/en/procedures/10/10-010-080-tr.html)

- Remove the customer accessory bracket.
- Remove the fuel supply and fuel drain lines. Refer to Procedure 006-024 in Section 6. (/qs3/pubsys2/xml/en/procedures/10/10-006-024.html)
- Remove the fuel filter. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/10/10-006-015-tr.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove two mounting capscrews from the IFSM.

Install two guide studs.

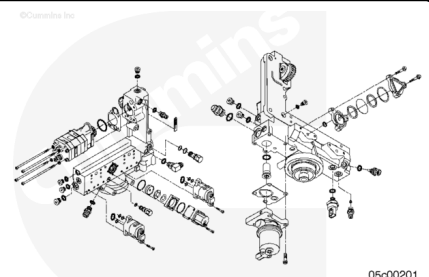
Remove the remaining capscrews, IFSM, and gasket. Discard the gasket, if damaged.

Disassemble

Automotive with CM870 and CM871

Remove all components:

- Sensors
- Regulators
- Check valve
- Fuel and timing actuators
- Fuel shut-off valve
- Gear pump (**only** remove the four long capscrews, as shown in the illustration.)
- Fuel inlet and drain fittings

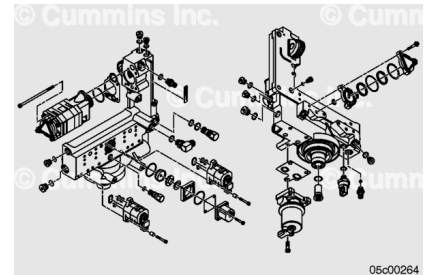


- Fuel dampers (**only** remove the two top capscrews of the damper assembly, as shown in the illustration.)
- All threaded plugs, o-rings, and gaskets.

CM570

Remove all components:

- Sensors
- Regulators
- Check valve
- Fuel and timing actuators
- Fuel shut-off valve
- Gear pump (**only** remove the four long capscrews, as shown in the illustration.)
- Fuel inlet and drain fittings
- Fuel dampers (**only** remove the two top capscrews of the damper assembly, as shown in the illustration.)
- All threaded plugs, o-rings, and gaskets.

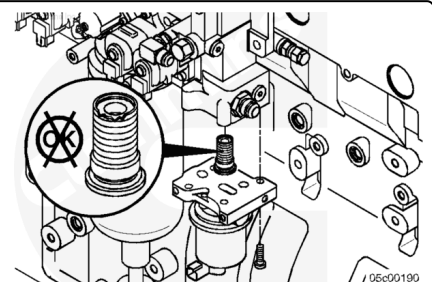


Clean and Inspect for Reuse

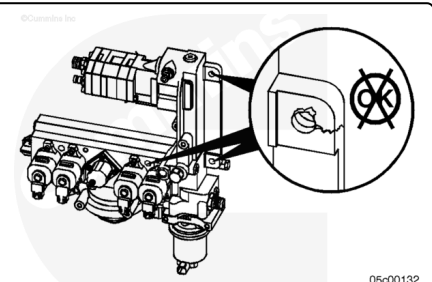
Automotive with CM870 and CM871

Inspect the anti-drain back valve for damage or improper sealing. Replace if necessary.

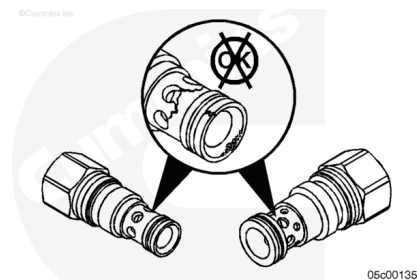
Verify that the anti-drain back valve seats properly and is free of damage or debris. If damage is found, replace the valve.



Inspect the mounting capscrew holes for cracks. Inspect the IFSM for leaks or cracks. Replace the module if cracks are found.

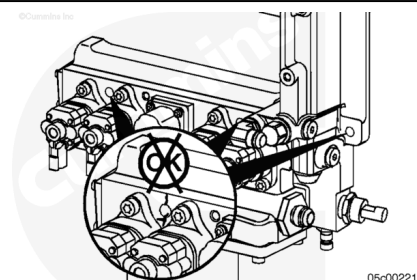


Inspect the 1724 kPa [250 psi] and the 2620 kPa [380 psi] fuel regulators for debris or damage (including the spring inside the regulators). Replace the regulator, if damage is found.



CM570

Inspect the mounting capscrew holes for cracks.

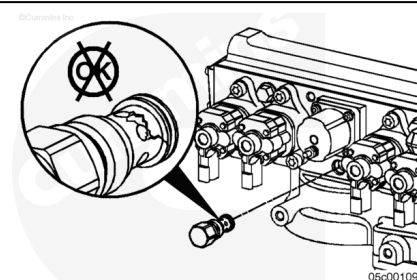


Remove the 1724 kPa [250 psi] fuel pressure regulator.

Inspect for debris or other damage. Replace the regulator, if necessary.

Install the fuel pressure regulator and tighten.

Torque Value: 27 n•m [239 in-lb]

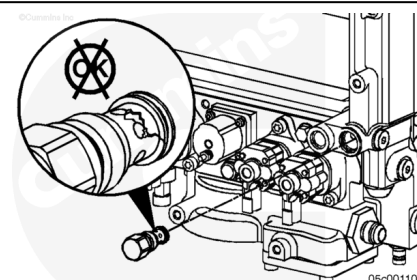


Remove the 2206 kPa [320 psi] fuel pressure regulator.

Inspect for debris or other damage. Replace the regulator, if necessary.

Install the fuel pressure regulator and tighten.

Torque Value: 27 n•m [239 in-lb]

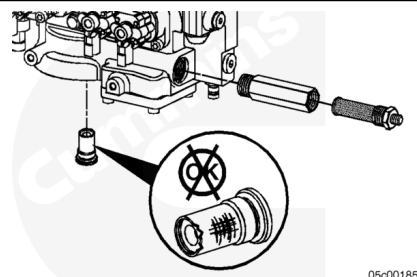


Remove the check valves located in the fuel inlet and fuel filter head.

Verify both seat properly and are free of damage or debris. Replace, if necessary.

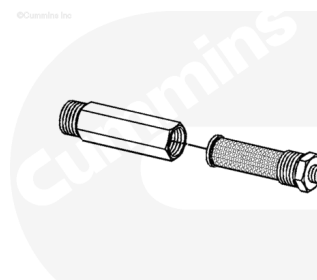
Install both check valves and tighten.

Torque Value: 17 n•m [150 in-lb]



Check the inlet filter screen for debris. Clean the filter, or replace, if necessary.

Install the inlet filter screen.



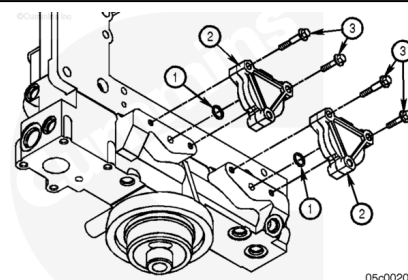
Assemble

Automotive with CM870 and CM871

Verify the seal (1) is in the groove of the fuel control damper (2) and add Lubriplate™ multi-purpose lubricant to hold the seal in place.

Attach the fuel control damper (2) with capscrews (3) and tighten.

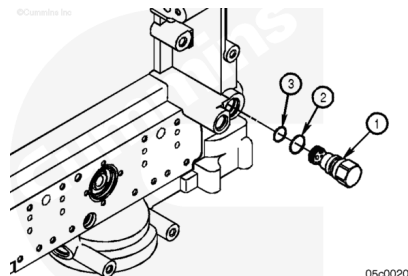
Torque Value: 9.5 n•m [84 in-lb]



Install the o-rings (2 and 3) onto the pressure regulator (1).

Install the (2620 kPa) [380 psi] pressure regulator (1) into the IFSM housing and tighten.

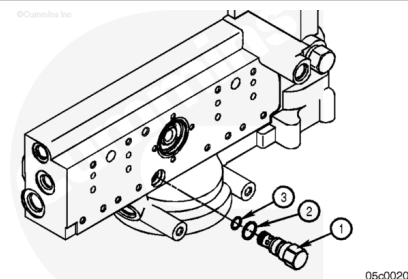
Torque Value: 27 n•m [239 in-lb]



Install the o-rings (2 and 3) onto the pressure regulator (1).

Install the (1724 kPa) [250 psi] fuel pressure regulator into the IFSM housing and tighten.

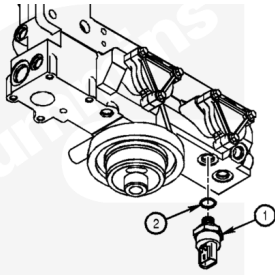
Torque Value: 27 n•m [239 in-lb]



Install the o-ring seal (2) onto the fuel pressure sensor (1).

Install the fuel pressure sensor (1) into the IFSM housing and tighten.

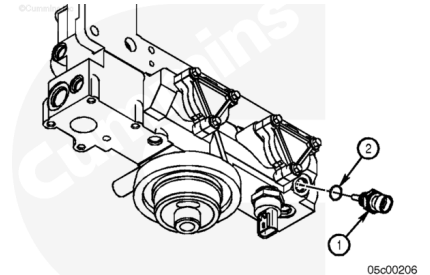
Torque Value: 14 n•m [124 in-lb]



05c00205

Install the o-ring seal (2) onto the fuel temperature sensor (1).
Install the fuel temperature sensor into the IFSM housing and tighten.

Torque Value: 23 n•m [204 in-lb]

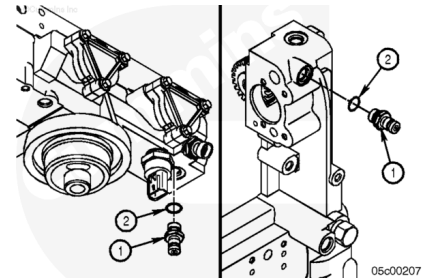


05c00206

Install the o-ring (2) onto the Compuchek™ fitting (1).

Install both Compuchek™ fittings (1) into the IFSM housing and tighten.

Torque Value: 9.5 n•m [84 in-lb]

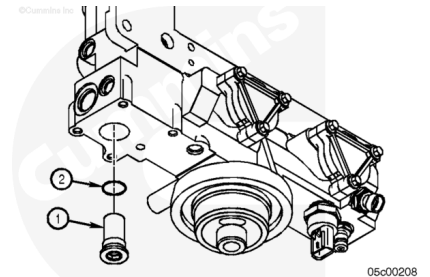


05c00207

Install the o-ring (2) onto the anti-drain back valve (1).

Install the anti-drain back valve (1) into the housing and tighten.

Torque Value: 17 n•m [150 in-lb]



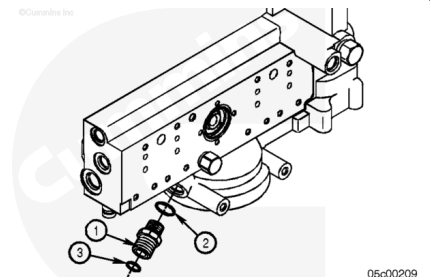
05c00208

Install the o-ring (2) onto the cooling plate fuel supply fitting (1).

Install the cooling plate fuel supply fitting (1) into the housing and tighten.

Torque Value: 27 n•m [239 in-lb]

Install the o-ring (3) onto the front face of the cooling plate fuel supply fitting.



05c00209

Install the o-ring (2) onto the fuel drain fitting (1).

Install the fuel drain fitting (1).

Turn the nut all the way to the top of the fitting.

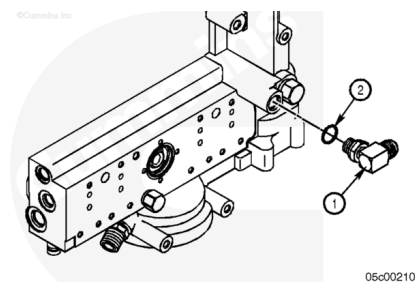
Thread the fitting all the way into the IFSM housing until it stops.

Turn the fitting back until the threaded end is oriented directly over the regulator.

Lock the fitting into position and tighten the locking nut.

Torque Value: 45 n•m [33 ft-lb]

Maintain the correct orientation.



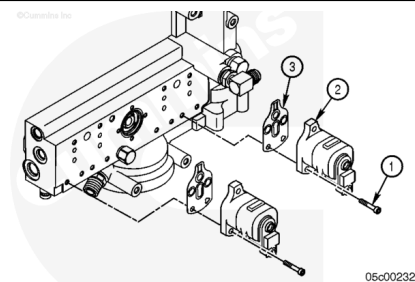
Use a small amount of Lubriplate™ multi-purpose lubricant in the grooves on the bottom of the metering actuators (2).

Install the actuator gasket (3) onto each actuator. Make sure to align each properly.

Install the actuators (2) onto the IFSM housing.

Use three capscrews (1) to mount each actuator to the housing and tighten the capscrews.

Torque Value: 15 n•m [133 in-lb]



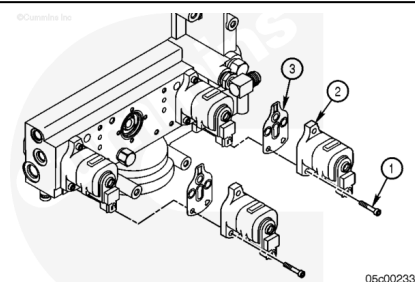
Use a small amount of Lubriplate™ multi-purpose lubricant in the grooves on the bottom of the timing actuators (2).

Install the actuator gasket (3) onto each actuator. Make sure to align each properly.

Install the actuators (2) onto the IFSM housing.

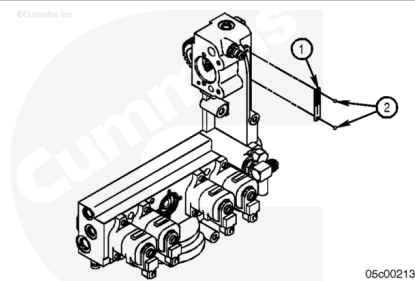
Use three capscrews (1) to mount each actuator to the housing and tighten the capscrews.

Torque Value: 15 n•m [133 in-lb]



Etch the same part number of the original IFSM on the data tag (1).

Attach the tag to the housing by pressing the screws (2) into the housing.



Install the seal (1) into the groove in the IFSM housing.

Place the actuator disc (2) over the seal (1), small diameter down.

Place the valve disc (3) onto the actuator disc (2) with the sealing surface down (spring pilot diameter up).

Place the valve spring (4) into the actuator housing (5) with the inside diameter on the valve disc pilot (3).

Place the actuator housing (5) over the actuator disc (2).

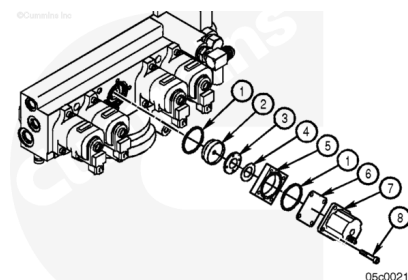
Install the seal (1) into the groove in the actuator housing (5).

Place the shutoff valve shield (6) over the actuator housing (5).

Align the hole pattern of the shutoff valve shield (6).

Place the solenoid (7) over the shutoff valve shield (6) taking care to align the holes and **not** pinch any seals. Orient the terminal facing the gear pump.

Use four capscrews (8) to attach the assembly to the housing.
Use an "x" pattern.



Torque Value: 5.5 n•m [49 in-lb]

Install the o-ring (2) onto the threaded plug (1).

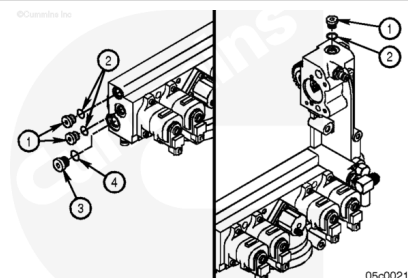
Install the three threaded plugs (1) into the housing.

Torque Value: 17 n•m [150 in-lb]

Install the o-ring (4) onto the large threaded plug (3).

Install the large threaded plug (3) into the housing.

Torque Value: 14 n•m [124 in-lb]



Install the o-rings (2) onto the threaded plugs (1).

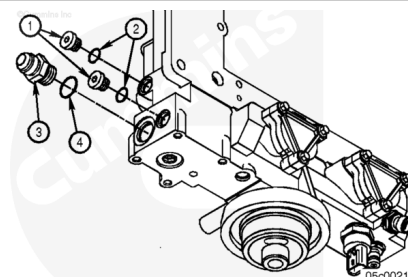
Install the two threaded plugs (1) into the housing.

Torque Value: 17 n•m [150 in-lb]

Install the o-ring (4) onto the fuel supply fitting (3).

Install the fuel supply fitting (3) into the housing.

Torque Value: 27 n•m [239 in-lb]

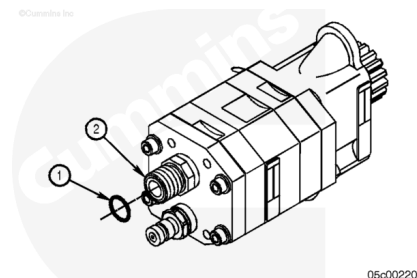


Install the o-ring (1) onto the fuel inlet fitting (2) from the ECM cooling plate.

Install the fitting (2) into the fuel gear pump and tighten.

Torque Value: 27 n•m [239 in-lb]

Install the o-ring (3) onto the fuel inlet fitting (2).



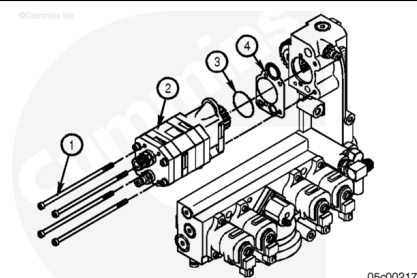
05c00220

Install the o-ring (3) onto the fuel gear pump (2).

Apply Lubriplate™ multi-purpose lubricant to the o-ring (3).

Install the gasket (4) onto the gear fuel pump.

Use four capscrews (1) to install the pump to the IFSM housing.



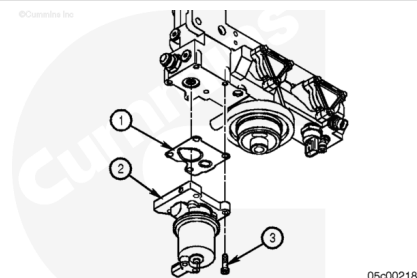
05c00217

Torque Value: 11 n•m [97 in-lb]

Install the lift pump gasket (1) onto the lift pump (2).

Use four capscrews (3) to install the lift pump to the housing.

Torque Value: 18 n•m [159 in-lb]



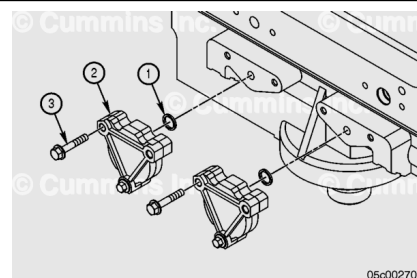
05c00218

CM570

Verify the seal (1) is in the groove of the fuel control damper (2) and add Lubriplate™ multi-purpose lubricant to hold the seal in place.

Attach the fuel control damper (2) with capscrews (3) and tighten.

Torque Value: 9.5 n•m [84 in-lb]

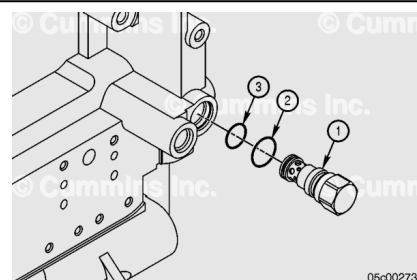


05c00270

Install the o-rings (2 and 3) onto the pressure regulator (1).

Install the 2206 kPa [320 psi] pressure regulator (1) into the IFSM housing and tighten.

Torque Value: 27 n•m [239 in-lb]

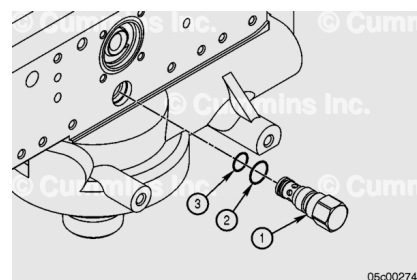


05c00273

Install the o-rings (2 and 3) onto the pressure regulator (1).

Install the 1724 kPa [250 psi] fuel pressure regulator into the IFSM housing and tighten.

Torque Value: 27 n•m [239 in-lb]

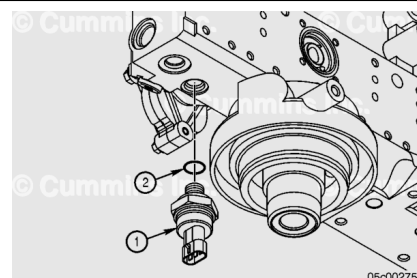


Install the o-rings (2) onto the fuel pressure sensor (1).

Install the fuel pressure sensor (1) into the IFSM housing and tighten.

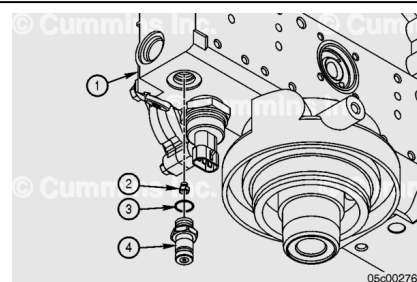
Torque Value: 14 n•m [124 in-lb]

Note : Some IFSMs contain an additional fuel pressure sensor.



Install the air bleed (2) onto the air bleed orifice into the IFSM (1) through the opening for the Compuchek™ fitting.

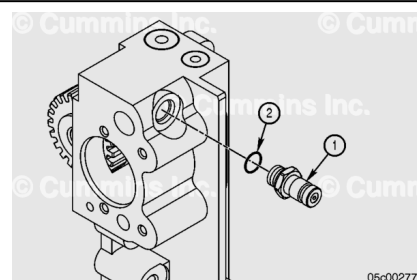
Torque Value: 7.9 n•m [70 in-lb]



Install the o-ring (2) onto the Compuchek™ fitting (1).

Install both Compuchek™ fittings (1) into the IFSM housing and tighten.

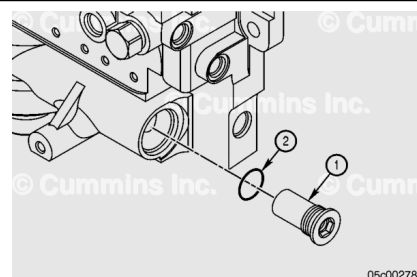
Torque Value: 9.5 n•m [84 in-lb]



Install the o-ring (2) onto the anti-drain back valve (1).

Install the anti-drain back valve (1) into the housing and tighten.

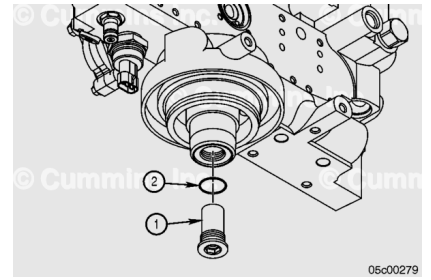
Torque Value: 17 n•m [150 in-lb]



Install the o-ring (2) onto the anti-drain back valve (1).

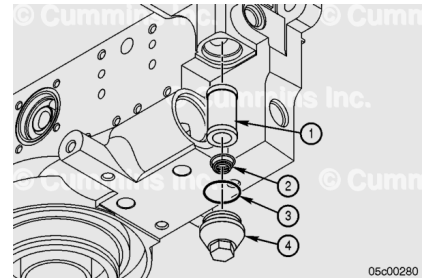
Install the anti-drain back valve (1) for the fuel filter into the housing and tighten.

Torque Value: 17 n•m [150 in-lb]



Install the filter screen (1), compression spring (2), o-ring (3), and filter cap (4) into the housing.

Torque Value: 17 n•m [150 in-lb]



Install the o-ring seal (2) onto the fuel drain fitting (1).

Install the fuel drain fitting (1).

Turn the nut all the way to the top of the fitting.

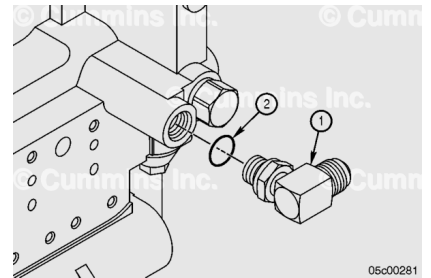
Thread the fitting all the way into the IFSM housing until it stops.

Turn the fitting back until the threaded end is oriented directly over the regulator.

Lock the fitting into position and tighten the locking nut.

Torque Value: 45 n•m [33 ft-lb]

Maintain the correct orientation.



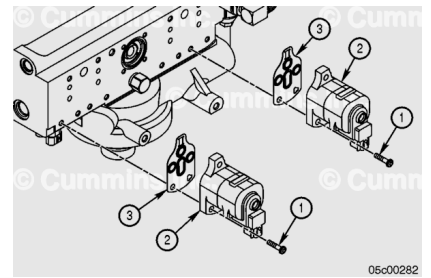
Use a small amount of Lubriplate™ multi-purpose lubricant in the grooves on the bottom of the metering actuators (2).

Install the actuator gasket (3) onto each actuator, making sure to align each properly.

Install the actuators (2) onto the IFSM housing.

Mount each actuator to the housing using three capscrews (1) and tighten the capscrews.

Torque Value: 15 n•m [133 in-lb]



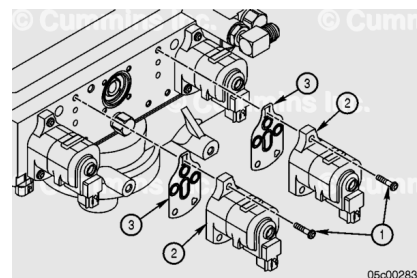
Use a small amount of Lubriplate™ multi-purpose lubricant in the grooves on the bottom of the timing actuators (2).

Install the actuator gasket (3) onto each actuator, making sure to align each properly.

Install the actuators (2) onto the IFSM housing.

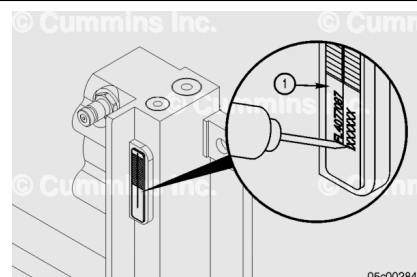
Mount each actuator to the housing using three capscrews (1) and tighten the capscrews.

Torque Value: 15 n·m [133 in-lb]



Etch the same part number of the original IFSM on the data tag (1).

Attach the tag to the housing by pressing the screws (2) into the housing.



Install the seal (1) into the groove in the IFSM housing.

Place the actuator disc (2) over the seal (1), small diameter down.

Place the valve disc (3) onto the actuator disc (2) with the sealing surface down (spring pilot diameter up).

Place the valve spring (4) into the actuator housing (5) with the inside diameter on the valve disc pilot (3).

Place the actuator housing (5) over the actuator disc (2).

Install the seal (1) into the groove in the actuator housing (5).

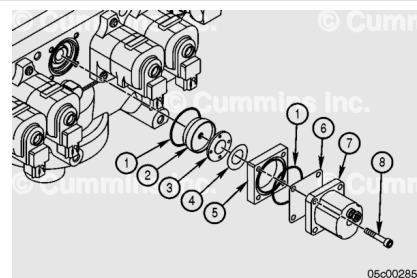
Place the shutoff valve shield (6) over the actuator housing (5).

Align the hole pattern of the shutoff valve shield (6).

Place the solenoid (7) over the shutoff valve shield (6), taking care to align the holes and **not** pinch any seals. Orient the terminal facing the gear pump.

Attach the assembly to the housing using four capscrews (8) and tighten using an "x" pattern.

Torque Value: 5.5 n·m [49 in-lb]



Install the o-ring (2) onto the threaded plug (1).

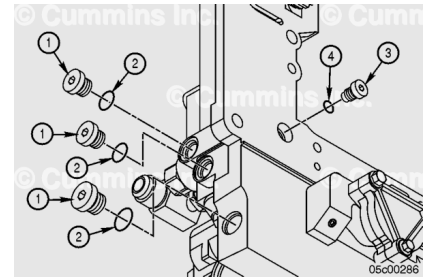
Install the three threaded plugs (1) into the housing.

Torque Value: 17 n·m [150 in-lb]

Install the o-ring (4) onto the large threaded plug (3).

Install the larger threaded plug (3) into the housing.

Torque Value: 14 n•m [124 in-lb]

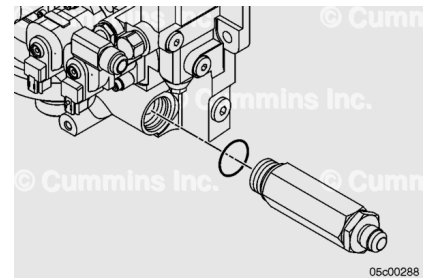


Install the o-ring onto the fuel supply fitting.

Install the fuel supply fitting into the housing.

Torque Value: 27 n•m [239 in-lb]

Install the o-ring onto the fuel inlet fitting.



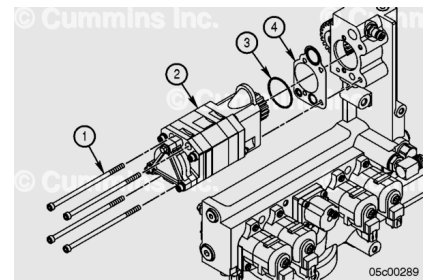
Install the o-ring (3) onto the fuel gear pump (2).

Apply Lubriplate™ multi-purpose lubricant to the o-ring (3).

Install the gasket (4) onto the gear fuel pump.

Use four capscrews (1) and install the pump to the IFSM housing.

Torque Value: 11 n•m [97 in-lb]

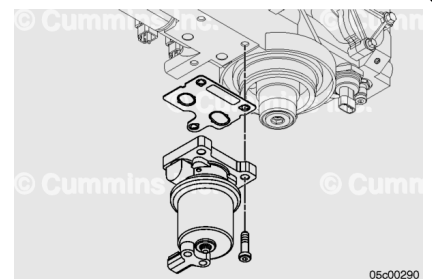


Install the lift pump gasket (1) onto the lift pump (2).

Install the lift pump to the housing using three capscrews (3).

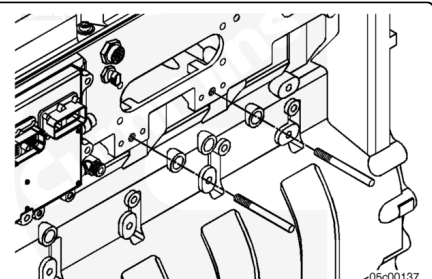
Note : Some early engines were not equipped with a lift pump. These engines use a hand priming pump or they have been fitted with an auxiliary lift pump.

Torque Value: 18 n•m [159 in-lb]

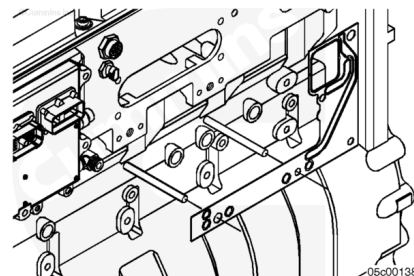


Install

Install two guide pins into the mounting holes on the cylinder head.

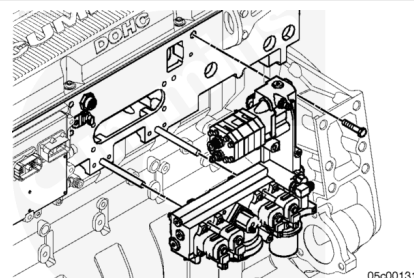


Install a new gasket onto the guide pins.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Install the IFSM on the guide pins.

Install and tighten the capscrews.

Remove the guide pins.

Install and tighten the remaining capscrews.

Torque Value: 45 n•m [33 ft-lb]

Finishing Steps

Automotive with CM870 and CM871

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the accessory bracket. It is located above the IFSM. Refer to Procedure 001-082 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-082.html)
- Install the ECM cooling plate fuel supply line (bottom) to the IFSM and ECM cooling plate fuel return line (top) to

the fuel pump. Refer to Procedure 006-006 in Section 6.
(/qs3/pubsys2/xml/en/procedures/10/10-006-006-tr.html)

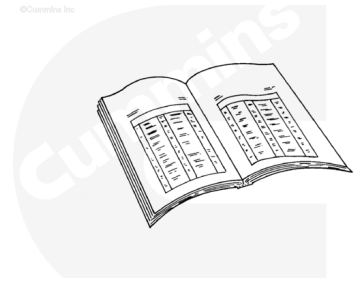
- Install the lift pump. Use the following procedure in the Troubleshooting and Repair Manual, CM870 Electronic Control System, Signature™ and ISX Engines, Bulletin 4021334. Refer to Procedure 019-396 in Section 19.
(/qs3/pubsys2/xml/en/procedures/70/70-019-396.html)
- Install the customer accessory bracket. Tighten to 18 N•m [159 in-lb].
- Connect the fuel supply and fuel drain lines. Refer to Procedure 006-024 in Section 6.
(/qs3/pubsys2/xml/en/procedures/10/10-006-024.html)
- Install the EGR differential pressure sensor adapter. Refer to Procedure 011-028 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-028.html)
- Install the EGR differential pressure sensor. Use the following procedure in the Troubleshooting and Repair Manual, CM870 Electronic Control System, Signature™ and ISX Engines, Bulletin 4021334. Refer to Procedure 019-370 in Section 19.
(/qs3/pubsys2/xml/en/procedures/70/70-019-370.html)
- Install the EGR differential pressure sensor tubes. Refer to Procedure 011-026 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-026.html)
- Connect the engine wiring harness to all temperature and pressure sensors, fueling and timing actuators, the fuel shutoff valve, the water-in-fuel sensor, and the fuel lift pump.
- Connect the air intake connection. Refer to Procedure 010-080 in Section 10.
(/qs3/pubsys2/xml/en/procedures/10/10-010-080-tr.html)
- Connect the EGR mixer to the air intake connection. Refer to Procedure 011-021 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-021-tr.html)
- Connect the battery cable. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/10/10-013-009.html)
- Install the fuel filter. Refer to Procedure 006-015 in Section 6. (/qs3/pubsys2/xml/en/procedures/10/10-006-015-tr.html)
- Prime the fuel system.

- Operate the engine to normal operating temperature and check for leaks.

CM570

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



ck800wa

- Install the fuel filter. Refer to Procedure 006-015 in Section 6. (</qs3/pubsys2/xml/en/procedures/10/10-006-015-tr.html>)
- Connect the fuel supply and fuel drain lines. Refer to Procedure 006-024 in Section 6. (</qs3/pubsys2/xml/en/procedures/10/10-006-024.html>)
- Install the customer accessory bracket. Tighten to 18 N•m [159 in-lb]).
- Connect the air intake connection. Refer to Procedure 010-080 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-080-tr.html>)
- Install the fuel control covers.
- Connect the engine wiring harness from all pressure and temperature sensors, fueling and timing actuators, the fuel shutoff valve, the water-in-fuel sensor, and the fuel lift pump.
- Connect the air inlet piping. Refer to Procedure 010-080 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-080-tr.html>)
- Connect the battery cables. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/10/10-013-009.html>)
- Operate the engine to normal operating temperature and check for leaks.

